CURRICULUM VITAE

Mohammad Reza Nasirzadeh

RADIO ASTRONOMY
INTERSTELLAR MEDIUM
MAGNETIC FIELDS

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01 RESEARCH INTERESTS

Multi-wavelength astronomy (radio/infrared/optical) · Interstellar medium physics and galactic structure · Cosmic ray propagation in galactic environments · Star formation in galactic and extra-galactic environments · Thermal/non-thermal processes in stellar and galactic systems · Molecular gas dynamics and dust properties in stellar formation · Magnetic fields in star formation regions and stellar evolution · Faraday tomography and polarization studies · Stellar populations and galactic evolution · Large survey data analysis and statistical methods

02 EDUCATION

06/2019 – 03/2022
M.Sc. in Gravity & Cosmology
University of Kharazmi

09/2014 – 06/2019
B.Sc. in Molecular Physics
University of Yazd

03 WORK EXPERIENCE

06/2020 – 04/2023
Research assistant
Institute For Research In Fundamental Sciences (IPM), Prof. F. S. Tabatabaei

04 RESEARCH EXPERIENCE

CURRENT PROJECT

Exploring Magnetic Fields in Molecular Clouds through Faraday Rotation and Faraday Tomography

Under guidance of: Dr. Mehrnoosh Tahani, Assistant Professor, University of South Carolina.
Related Collaboration: GMIMS (Global Magneto-Ionic Medium Survey) international collaboration.

M.SC. **Research Assistant, Studying Radio-IR Correlation in IC 342 based on LoTSS Data**

School of Astronomy, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran
Supervisor: Dr. F. S. Tabatabaei, School of Astronomy, IPM, Tehran, Iran
Related Collaboration: LOFAR Key Science Projects (Magnetic Key Science Project), NEARBY GALAXIES group.

B.SC. Analysis and Reduction of Data Taken from the Giant Meteor Wave Radio Telescope (GMRT)

Research on Phenomenon related to the Very Low Frequency (VLF) Band of Radio Electromagnetism Spectrum, including:

Solar Flares and Sudden Ionospheric Disturbances (SIDs) | Meteor | Meteorological Phenomenon | Ionospheric Anomalies | Earthquake Prediction

Designing and Constructing Small Radio Detectors to Study the Phenomenon in VLF Band

Supervisor: Dr. Zahra Rezaei, Yazd University, Yazd, Iran

05 PUBLICATIONS

2024 **Radio–FIR correlation: A probe into cosmic ray propagation in the nearby galaxy IC 342**

Nasirzadeh, M.R., Tabatabaei, F.S., Beck, R., Heesen, V., Howaida, P., Reina-Campos, M., Paladino, R., Dettmar, R.-J., Chyzy, K.T. — *Astronomy & Astrophysics*

SUBMITTED **Properties of giant H ii regions of M33 at low radio frequencies: implications for spectral energy distributions of galaxies**

Co-author — submitted

06 OBSERVATIONAL DATA EXPERIENCE

- Radio instruments and surveys: LOFAR, GMIMS, CHIME, and VLA.
- Planck's dust polarization data analysis.
- UV data processing with GALEX.
- IR/FIR analysis using Spitzer, Herschel, and WISE.
- 3D interstellar extinction Profile analysis.
- HI spectral line data analysis.
- + Other dataset (H-alpha/Gamma/Extinction/CO)

07 SELECTED COMPUTER SKILLS

OPERATING SYSTEMS Windows, Linux

ANALYSIS & PROGRAMMING AIPS, DS9, Python (NumPy, AstroPy and...)

08 PROJECTS

Astra CREATOR AND MAINTAINER · ONGOING

Open-source tool that finds academic positions and ranks potential supervisors by an explainable 0–100 fit from NASA ADS, OpenAlex and arXiv, running locally as a CLI, a web dashboard and an offline desktop app.

Python | Next.js | FastAPI | Tauri mimdot.github.io/projects/astra

09 SCHOLARSHIPS, HONOURS AND AWARDS

- IPM annual grant in Astronomy and Astrophysics for research project "Detailed Study of Radio-IR correlation in IC 342" (2019-2022).
- Top 10% in MSc University exam entrance (2019).
- Top 5% among Gravity And Cosmology MSc students at Kharazmi University (2019-2022).

10 SCIENCE OUTREACH

- Voluntary Astronomy, Geography and History teacher for refugee children (2019).
- Voluntary Literature teacher for refugee children (2018-2019).
- Voluntary Scientific editor for different independent Astronomy and Science groups (2016-2022).
- Head of Student Astronomical Society of Yazd University (2017-2018).

11 REFERENCES

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